

Type 2 diabetes prevention across the life course

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Type 2 diabetes (T2D) prevention efforts have largely focused on intervening when dysglycemia is already established. We propose that T2D prevention be reframed around prediabetes remission, with preservation and restoration of normoglycemia as the optimal clinical goal. The transition from normoglycemia through increasing dysglycemia to T2D is progressive and cumulatively shaped by biological, behavioral and environmental exposures across the life course. Prediabetes (intermediate hyperglycemia) remission is an achievable, pragmatic and measurable prevention target. Here we provide a life-course risk architecture for T2D integrating developmental, transitional and contextual determinants, defining critical windows of amplified metabolic vulnerability and potential restoration of normoglycemia. Precision prevention should target mechanistic heterogeneity, with aligned interventions that remain scalable, affordable and adaptable across socioeconomic settings. Our framework identifies ten priorities in T2D prevention, moving beyond traditional approaches toward context-specific, actionable interventions capable of altering the natural history of disease early in the life course and restoring metabolic health.

In 2024, an estimated one in nine adults (~589 million) globally were living with diabetes, with the prevalence projected to rise to 853 million in 2050¹, and roughly US\$1 trillion of global health expenditures were spent on diabetes in 2021². Around 90–95% of current cases are type 2 diabetes (T2D), which has multiple modifiable and non-modifiable risk factors and is often associated with excess adiposity and ectopic lipid deposition attributed to poor dietary intake, low amounts of physical activity and unfavorable social determinants of health. Public health approaches to reduce the prevalence and incidence of obesity, T2D and associated cardiometabolic consequences largely comprise

recommendations for lifestyle modifications, such as improving the consumption of healthy foods, increasing physical activity, promoting good sleep hygiene and smoking cessation. Despite these widespread recommendations by public health bodies such as the World Health Organization (WHO) and national health agencies^{3,4}, the prevalence of T2D has continued to rise over the past four decades.

The rising prevalence of T2D, global inequitable distribution of risk and access to preventative interventions, and mechanistic heterogeneity among those at high risk of developing T2D illustrate that the risk for T2D is dynamic, graded and cumulatively shaped by exposures across

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the life course. A prevention strategy that is intentionally delayed until the latter stages of dysglycemia—defined solely by glycemic thresholds—systematically misses earlier opportunities for intervention, when risk is emerging and the trajectory toward diabetes is most readily reversed.

In this Perspective, we contend that the prevention of T2D is feasible, with the success of preventative interventions being enhanced by accounting for etiological heterogeneity, risk factors across the life course and contextual drivers such as social determinants of health. We provide a life-course risk architecture for T2D, focusing on key developmental and transitional periods in which metabolic vulnerability is programmed and potentially mitigated. We also highlight strategies for individuals with prediabetes (intermediate hyperglycemia) or at high risk of developing T2D (such as women who have experienced gestational diabetes mellitus (GDM)), to facilitate the prevention of overt disease. We propose an approach that explicitly recognizes the cumulative lifetime risk of T2D. The goal is prediabetes remission, incorporating precision approaches that leverage the heterogeneity of T2D and the pathophysiology that precedes the onset of established disease. This will not only prevent or slow the development of T2D, but also avert the complications associated with it.

Prediabetes (intermediate hyperglycemia) and T2D

T2D is defined as persistently elevated glycemia (based on various thresholds for plasma glucose). Glycemia levels that fall below these thresholds but remain higher than normoglycemic values are commonly termed prediabetes or intermediate hyperglycemia (Box 1). The term intermediate hyperglycemia (or nondiabetic hyperglycemia, commonly used in the UK), reflects the view that this state represents part of a glycemic continuum toward T2D rather than a distinct, pre-disease entity, since glycemic states exist on a spectrum and confer a high short-term risk of progression to T2D⁵. Impaired fasting glucose (IFG) and impaired glucose tolerance (IGT) are often grouped together as prediabetes or intermediate hyperglycemia, yet they reflect partially distinct metabolic phenotypes and pathogenic mechanisms. Most individuals present with either IFG or IGT⁶. However, individuals with combined IFG and IGT typically exhibit more severe disturbances across both hepatic and peripheral insulin sensitivity and β -cell function and are at higher cardiometabolic risk than those with IGT alone^{7,8}.

Although plasma glucose-based thresholds are helpful for the clinical classification of dysglycemia, relying on these alone has substantial limitations for the effective prevention of T2D. Threshold-based approaches can obscure population-specific risk (which may develop at lower adiposity or through adiposity-independent mechanisms) and can delay action by treating risk as a binary state rather than a cumulative, graded continuum. Plasma-glucose-based thresholds do not directly represent cumulative exposure to upstream drivers, including social disadvantage, genetic and epigenetic determinants, early-life programming, pregnancy-related metabolic stressors and intergenerational effects. They also fail to capture other biochemical abnormalities, such as low-grade inflammation, lipids and liver enzyme abnormalities, that may coexist with hyperglycemia and contribute to T2D risk, increasing the chances that distinct risk categories and life-course vulnerabilities may be overlooked. Therefore, understanding the mechanisms and pathways underlying chronic hyperglycemia is crucial for prevention.

Pathways and complications of prediabetes (intermediate hyperglycemia)

Dysglycemia evolves along a continuum that often begins years before prediabetes (intermediate hyperglycemia) is clinically recognized, reflecting variable contributions of insulin resistance, β -cell dysfunction, ectopic lipid deposition and low-grade inflammation^{9,10}. Mechanistic frameworks differ according to the initiating lesion: the

BOX 1

Definitions of prediabetes (intermediate hyperglycemia) and T2D

T2D is a diagnosis reached when other forms of diabetes have been excluded through specific diagnostic tests. For example, in young people, early-stage type 1 diabetes needs to be excluded, considering the different course of this disease and disease-modifying therapies that are emerging²⁰³. Rare genetic forms of diabetes that present with dysglycemia (for example, monogenic diabetes) need to be identified, and rapidly evolving, insulin-requiring forms of diabetes need to be diagnosed early to guide the therapeutic approach.

T2D is defined as persistently elevated glycemia, diagnosed when any one of following criteria is met: an FPG of $\geq 7.0 \text{ mmol l}^{-1}$ (126 mg dl^{-1}), an HbA1c of $\geq 6.5\%$ (48 mmol mol^{-1}), a 2 h plasma glucose of $\geq 11.1 \text{ mmol l}^{-1}$ (200 mg dl^{-1}) after a 75 g oral glucose tolerance test, or a random plasma glucose of $\geq 11.1 \text{ mmol l}^{-1}$ (200 mg dl^{-1}) accompanied by symptoms of high glucose levels.

Prediabetes represents an intermediate glycemic state between normoglycemia and diabetes, defined by specific glucose threshold criteria according to international guidelines. The WHO and IDF use the term intermediate hyperglycemia, reflecting the view that this state represents part of a glycemic continuum toward T2D rather than a distinct, pre-disease entity. Intermediate hyperglycemia includes IFG (an FPG of $6.1\text{--}6.9 \text{ mmol l}^{-1}$ ($110\text{--}125 \text{ mg dl}^{-1}$)) or IGT (a 2 h post-load glucose of $7.8\text{--}11.1 \text{ mmol l}^{-1}$ ($140\text{--}199 \text{ mg dl}^{-1}$))²⁰⁴. The ADA, using the term prediabetes, adopts broader criteria, including a lower IFG cut-off (an FPG of $5.6\text{--}6.9 \text{ mmol l}^{-1}$ ($100\text{--}125 \text{ mg dl}^{-1}$)) and an HbA1c range of $5.7\text{--}6.4\%$ ($39\text{--}47 \text{ mmol mol}^{-1}$)²⁰⁵. By contrast, the UK's National Institute for Health and Care Excellence guidelines use a slightly higher HbA1c threshold of $6.0\text{--}6.4\%$ ($42\text{--}47 \text{ mmol mol}^{-1}$), whereas its IFG and IGT criteria align with those of the WHO²⁰⁶. The IDF published a Position Statement proposing the 1 h post-load plasma glucose with a validated cut-off point of 8.6 mmol l^{-1} (155 mg dl^{-1}) as a more sensitive and practical method to screen for intermediate hyperglycemia in people at risk²⁰⁷.

Notably, these definitions overlap but are not identical (particularly for fasting glucose), which can meaningfully shift the proportion of populations labeled at risk of T2D and its complications. It is also important to acknowledge that reliance on HbA1c alone is not sufficient for the determination of intermediate hyperglycemia; many cases of IGT are missed without the use of an oral glucose tolerance test, leading to missed opportunities for prevention and intervention. Reflecting a shift from dichotomous definitions, risk stratification using a staging schema for early diagnoses of prediabetes was proposed²⁰⁸. Further refinement of the staging schema will be required to accurately reflect populations at high risk and simplify categorization, making this applicable in clinical practice.

prevailing model invokes obesity-related insulin resistance with compensatory hyperinsulinemia that ultimately unmasks limited β -cell reserve, whereas alternative paradigms propose primary hyperinsulinemia leading to insulin resistance or primary β -cell failure with relative hypoinsulinemia^{11,12}. These mechanisms may manifest as distinct but also overlapping glycemic phenotypes: IFG is dominated by hepatic insulin resistance and increased endogenous glucose production, amplified by non-esterified fatty acid-driven lipotoxic

signaling, whereas IGT predominantly reflects skeletal muscle insulin resistance with defective insulin-stimulated glucose disposal^{13,14}. With progression, glucotoxicity, mitochondrial dysfunction and disrupted inter-organ communication converge to drive systemic subclinical inflammation, which further impairs insulin signaling and secretion. These biological mechanisms underlying prediabetes (intermediate hyperglycemia) and its relationship with long-term T2D and complications support early detection—a central tenet of national and international diabetes strategies aimed at preventing complications. However, any level of persistent hyperglycemia is associated not only with an increased risk of developing T2D but also with an increased risk of microvascular and macrovascular complications^{15,16}.

Early microvascular changes—particularly diabetic neuropathy and retinopathy—have been observed at plasma glucose and HbA1c levels within the prediabetic range. Population-based studies demonstrate a graded increase in retinopathy prevalence with rising fasting plasma glucose (FPG) and HbA1c, with some retinal lesions detected below diagnostic thresholds for diabetes^{17,18}. The American Diabetes Association (ADA) prediabetes criteria (Box 1) appear to be more sensitive for identifying individuals at risk of early retinopathy¹⁹, supporting the concept that microvascular injury begins before overt T2D. Epidemiological analysis of baseline data from the ePREDICE study showed that, in people with prediabetes, the prevalences of kidney dysfunction, hyperfiltration, albuminuria and chronic kidney disease stage G3a were 9.2%, 2.0%, 4.7% and 3.0%, respectively²⁰.

Macrovascular complications also represent a major source of morbidity and mortality in prediabetes (intermediate hyperglycemia). IFG and IGT are associated with increased risks of cardiovascular disease, including coronary heart disease, stroke, heart failure and atrial fibrillation, as well as higher all-cause and cardiovascular mortality^{21,22}. However, after adjustment for other cardiovascular risk factors, IFG was no longer an independent predictor of mortality (DECODE study)²³. Beyond classical vascular outcomes, prediabetes (intermediate hyperglycemia) is linked to a broad spectrum of adverse health outcomes, including dementia and malignancy²⁴. Meta-analyses indicate increased risks of total cancer, hepatocellular carcinoma and breast cancer, along with non-cancer and non-vascular causes of death such as kidney disease and infections^{21,24}. Prediabetes (intermediate hyperglycemia) often coincides with metabolic dysfunction-associated steatotic liver disease and liver fibrosis^{25,26} and cardiovascular–kidney–metabolic syndrome²⁷. Therefore, a key priority for T2D prevention (Box 2 provides others) should be prevention trials and prospective studies that move beyond glycemic endpoints and evaluate cardiovascular and renal metrics, mortality and other clinically meaningful and emerging outcomes, including cancer incidence.

Prediabetes remission

To prevent T2D and subsequent complications, current strategies primarily focus on slowing the disease progression to T2D in individuals with prediabetes (intermediate hyperglycemia). Remission of prediabetes represents a conceptual shift in prevention strategies by targeting a return to normoglycemia^{28–31}. This approach offers the opportunity (at any point in the life course) to preserve β -cell function and normalize ectopic lipid deposition and insulin sensitivity before irreversible decline occurs, and to restore metabolic health²⁸. Although many guidelines recommend multimodal lifestyle interventions that target 5–10% weight loss for high-risk groups⁵, they do not recommend glycemic targets, including normoglycemia. Moreover, relative percentage reductions are not typically used as the primary therapeutic goal in hypertension or dyslipidemia; in both conditions, fixed clinical targets are pursued for well-established reasons³². Importantly, there is also no hard evidence demonstrating that a 5–10% weight loss target is optimal for diabetes prevention compared with smaller or larger amounts of weight loss. Adding targeted glycemic goals to weight loss goals enhances protection against T2D progression and

BOX 2

Ten priorities for T2D prevention

1. Move beyond thresholds toward biologically grounded staging.

Prediabetes is a misnomer for a continuous and biologically heterogeneous process of metabolic decline. Early T2D should be defined by pathophysiological risk and trajectory, not by glucose values alone.

2. Redesign screening for the life course.

Waiting until midlife hyperglycemia emerges is no longer sufficient. Screening should incorporate weight trajectories, gestational dysglycemia, family history and emerging biomarkers rather than relying largely on age and BMI-based criteria.

3. Recognize heterogeneity within dysglycemia.

Dysglycemia is not a uniform metabolic state. IFG and IGT are clinical expressions of broader biological heterogeneity and should not automatically trigger identical prevention strategies.

4. Make remission durable.

Remission must reflect biological disease modification, not merely temporary glycemic normalization. Predictive markers are needed to identify who can achieve remission, who cannot and who is at risk of relapse.

5. Prevention must not wait for abnormal glucose values.

Maintenance of metabolic mobility—preserved insulin sensitivity, β -cell adaptability, weight stability and metabolic flexibility—should itself become a primary prevention target.

6. Intervene when biology is most plastic.

Critical windows for prevention extend from preconception and pregnancy to childhood and adolescence. Context-specific strategies should reflect the life-course architecture of risk.

7. Build dynamic precision prevention tools.

Artificial intelligence and multimodal data integration can support adaptive risk models incorporating genetics, omics, life events and environmental exposures. The goal is to identify modifiable inflection points.

8. Define the role of pharmacotherapy in prevention.

Potent incretin-based and other emerging therapies may have a role in selected high-risk groups. Prevention must avoid both therapeutic inertia and indiscriminate medicalization.

9. Demand evidence for hard clinical outcomes.

Prevention trials must move beyond glycemic endpoints to assess cardiovascular events, kidney disease, other diabetes-associated disorders, premature mortality and healthy lifespan.

10. Embed equity at the core.

Diabetes prevention must not become privilege based. Scalable, affordable and culturally appropriate strategies are essential, and individuals and communities at increased risk should be involved in prevention design.

reduces T2D incidence for years after the intervention is stopped^{29,33}. Prediabetes remission can arise from lifestyle-induced weight loss, pharmacological therapy, bariatric surgery or, in some cases, spontaneous metabolic improvement^{31,34}. Remission is also possible with lifestyle interventions without weight loss^{35,36}. Importantly, remission also challenges threshold-based thinking: if returning from dysglycemia

Table 1 | Lifestyle interventions and clinically available glucose-lowering pharmacotherapies evaluated for T2D prevention

Agent	Study	Number of participants	Baseline characteristics	Treatment duration	Rate of achieving prediabetes remission	Effect on new-onset T2D compared with control group
Lifestyle intervention studies						
Diet and exercise	DaQing IGT and Diabetes Study ¹⁹⁰	577	Men and women with IGT	6 years	13%	RRR: diet 33%; exercise 47%; diet + exercise 38%
	Finnish DPS ¹⁹¹	522	Men and women with IGT	3.2 years	–	RRR: 58%
	US DPP ^{37,192}	3,234	Men and women with IGT or IFG	2.8 years	52% across 3 years	RRR: 58%
	PLIS ^{29,35}	1,160	Men and women with IGT or IFG	1 year	34%	–
	Indian DPP ¹⁹³	531	Men and women with IGT	2.5 years	–	RRR: 29%
Exercise	IGT trial, Japan ¹⁹⁴	458	Men and women with IGT	4 years	–	RRR: 67%
Pharmacological intervention studies						
Metformin	US DPP ^{37,192}	3,234	Men and women with IGT or IFG	2.8 years	52%	RRR: 31%
	Indian DPP ¹⁹³	531	Men and women with IGT	2.5 years	–	RRR: 26%
Tirzepatide	SURMOUNT ¹⁹⁵	1,032	Men and women with both obesity and IGT or IFG	176 weeks	Up to 93.3% in intervention group; 58.9% in control group	HR: 0.07 (95% CI: 0–0.10; $P < 0.001$)
Liraglutide	SCALE ¹⁹⁶	2,254	Men and women with IGT or IFG and a BMI of at least 30 kg m ⁻² , or at least 27 kg m ⁻² with comorbidities	56 weeks	–	RRR: 79%
Semaglutide	STEP 10 ¹⁹⁷	207	Men and women with a BMI of ≥ 30 kg m ⁻² and IGT or IFG	52 weeks	81% in intervention group; 14% in control group	–
Glimepiride	NANSY ¹⁹⁸	274	Men and women with IGT	3.7 years	–	No significant effect on diabetes risk reduction
Pioglitazone	ACT NOW ¹⁹⁹	602	Men and women with IGT	Median follow-up 2.4 years	–	HR: 0.28 ($P < 0.005$)
Acarbose	STOP-NIDDM ²⁰⁰	714	Men and women with IGT	3.3 years	–	RRR: 25%
	ACE ²⁰¹	6,522	Men and women with coronary heart disease and IGT	6.6 years	–	IRR: 0.82 (95% CI: 0.71–0.94; $P = 0.005$)
Insulin glargine	ORIGIN ²⁰²	12,537 (including 1,452 without pre-existing T2D)	Men and women with CVD risk factors plus IFG, IGT or T2D	Median follow-up 6.2 years	–	OR: 0.72 (95% CI: 0.58–0.91; $P = 0.006$)

ACE, Acarbose Cardiovascular Evaluation; ACT NOW, Actos Now for Prevention of Diabetes; CI, confidence interval; CVD, cardiovascular disease; DPS, Diabetes Prevention Study; HR, hazard ratio; IRR, incidence rate ratio; NANSY, NEPI Antidiabetes Study; OR, odds ratio; ORIGIN, Outcome Reduction with an Initial Glargine Intervention; RRR, relative risk reduction; SCALE, Satiety and Clinical Adiposity Liraglutide Evidence in Nondiabetic and Diabetic Subjects; STEP, Semaglutide Treatment Effect in People with Obesity; STOP-NIDDM, Study to Prevent Non-Insulin-Dependent Diabetes Mellitus.

to normoglycemia modifies risk, prediabetes (intermediate hyperglycemia) should not be regarded as a fixed diagnostic category but as a dynamic, potentially reversible stage within a broader biological trajectory toward T2D. This concept supports a pathophysiology-based staging model in which intervention intensity follows biological risk, remission potential and trajectory, rather than glucose values alone—another central priority for T2D prevention (Box 2).

Studies such as the Diabetes Prevention Program (DPP)^{36–38} and Prediabetes Lifestyle Intervention Study (PLIS)^{29,35} demonstrated that approximately 34–52% of adults with prediabetes can achieve remission through substantial body weight loss via increased physical activity and diet modification³⁹ (Table 1). Among individuals who experienced at least 5% weight loss through lifestyle intervention, those who achieved remission of prediabetes exhibited a larger reduction in visceral adipose tissue volume and greater improvements in insulin sensitivity than those who did not experience remission^{29,37}. However, the magnitude

of weight loss and accompanying reductions in hepatic fat content and skeletal muscle fat mass were similar across both groups. The PLIS also revealed that attainment of normoglycemia with lifestyle intervention but without weight loss was associated with a shift from visceral to subcutaneous fat depots, even with mild weight gain³⁵. Improvements in insulin sensitivity, along with enhanced β -cell function and β -cell glucagon-like peptide-1 (GLP-1) sensitivity, appear to account for this pattern of remission³⁵. Furthermore, individuals with elevated 1 h plasma glucose, a marker of early-stage dysglycemia, show particularly strong insulin sensitivity and β -cell function remission potential⁴⁰. These findings broaden the mechanistic understanding of prediabetes remission and highlight body fat redistribution, enhanced insulin sensitivity and β -cell function as central therapeutic goals^{28,31}.

Prediabetes remission is associated with a reduced incidence of T2D, with effects that persist beyond the active intervention period. In the PLIS, people who achieved the recommended 5% weight loss

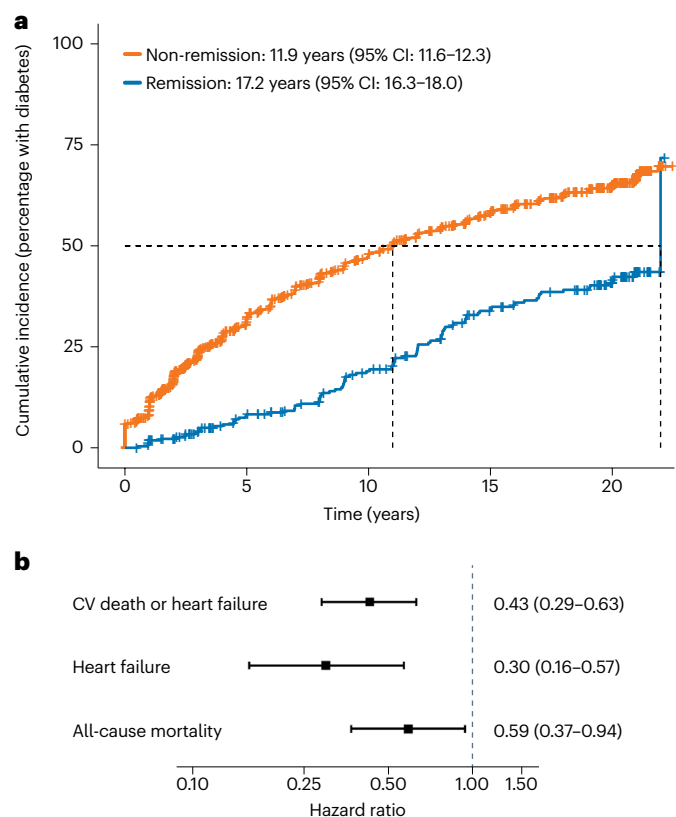


Fig. 1 | Long-term cardiometabolic benefits associated with prediabetes remission. **a**, Cumulative incidence of T2D over long-term follow-up (22 years) according to remission status. Participants who achieved prediabetes remission had a delayed progression to T2D and a longer restricted mean survival time without diabetes (by 5.2 years; 95% confidence interval: 4.3–6.1) compared with those without remission. **b**, Hazard ratios for cardiovascular outcomes comparing participants with prediabetes remission versus non-remission. Prediabetes remission was associated with lower risks of cardiovascular (CV) death or heart failure (composite endpoint), heart failure and all-cause mortality. The error bars and numbers in parentheses represent 95% confidence intervals. Data from ref. 36, research conducted as part of the DPPOS.

and experienced prediabetes remission compared with those who did not showed a 73% lower incidence of T2D incidence two years after the intervention³³. The DPP Outcome Study (DPPOS) demonstrated a 56% reduced risk of incident T2D over 10 years among those who achieved normal fasting or 2 h glucose at least once³⁰. In the DPPOS, participants who achieved remission after one year remained free of diabetes for an average of 5.2 years longer over the 22-year follow-up than those who did not achieve remission³⁶ (Fig. 1a). Prediabetes remission also confers broad cardiometabolic advantages, including a reduced incidence of both microvascular and major cardiovascular complications. In the PLIS, remission was linked to an improvement in markers of kidney function, such as a reduced albumin-to-creatinine ratio and enhanced small-vessel integrity in the skin⁴¹. Long-term data also showed that prediabetes remission is associated with a lower prevalence of microvascular disease, including nephropathy and retinopathy⁴¹. Moreover, a joint analysis of the DPPOS and Da Qing Diabetes Prevention Study indicated that prediabetes remission is associated with an approximately 44–89% reduction in composite cardiovascular events. A pooled meta-analysis of data from the DPPOS and Da Qing Diabetes Prevention Study yielded significant reductions in the risk of cardiovascular death or hospitalization for heart failure and all-cause mortality years after the intervention stopped³⁶ (Fig. 1b). The observed benefits of remission held after inverse probability weighting for baseline differences, suggesting that differential risk at baseline was not the major driver

of the observed benefits. Instead, these legacy effects (that is, where protection outlasts the intervention period) are mediated by improved metabolic health, including reduced glucotoxicity, improved insulin sensitivity and secretion, altered body fat distribution and lower low-grade inflammation. Emerging evidence further suggests that prediabetes remission may also be associated with a lower incidence of certain cancers^{42,43}.

There is no consensus on whether pharmacotherapy should be used for diabetes prevention. However, given the limited durability of lifestyle interventions alone, pharmacological agents may serve as useful adjuncts to achieve regression of prediabetes (intermediate hyperglycemia) in selected high-risk populations (Box 3).

Life-course risk architecture

Given the cumulative exposures and heterogeneity of pathophysiological pathways leading to T2D, it is essential to adopt a cumulative life-course approach to identify high-risk individuals for the correct preventive interventions at the right time (Fig. 2). Across the life course, metabolic vulnerability is also shaped by individual- and community-level social and environmental exposures—including income, occupation, social stress, housing and neighborhood conditions, and access to healthcare—through their effects on stage-specific lifestyle and biological factors^{44,45}.

Preconception

The parents' metabolic health at the time of conception plays a decisive part in programming the long-term health trajectory of their offspring, which substantially impacts an individual's risk of developing T2D in later life⁴⁶. This intergenerational vicious cycle is best articulated by the developmental origins of health and disease concept^{47–49}. Genetic susceptibility and gene–environment interactions may further shape this intergenerational risk, with early development representing a particularly sensitive period for epigenetic adaptation⁵⁰. Although DNA methylation changes can occur later in life in response to environmental exposures, their causal role in T2D risk remains less clearly established^{51,52}.

Having a parent with T2D is a well-established risk factor for developing T2D⁵³. Although genetics plays a part, sibling-pair analyses demonstrate that preconception maternal hyperglycemic status notably increases offspring risk of T2D and obesity. Siblings born after their mother was diagnosed with T2D have a substantially higher body mass index (BMI) in adolescence and a higher risk of T2D up to early adulthood compared with their siblings born before her diagnosis⁵⁴. Similar findings were reported in an offspring cohort study with Canadian First Nations participants, in which prenatal exposure to maternal hyperglycemia accelerated the onset of T2D in offspring⁵⁵. Paternal factors at conception are also influential on diabetes outcomes in offspring. Experimental studies show that factors such as circadian rhythm disruption or a short-term high-fat diet in fathers can also alter offspring metabolism through non-genetic pathways, including stress-mediated signaling at conception and modifications in sperm mitochondrial transfer RNA^{56,57}.

Pregnancy

Pregnancy is a critical window during which maternal metabolic status influences long-term health outcomes in both offspring and mothers. Exposure to maternal hyperglycemia, even below diagnostic thresholds for GDM, is consistently linked to adverse metabolic profiles in offspring^{58,59}. Follow-up of the Hyperglycemia and Adverse Pregnancy Outcome Study demonstrated a graded, linear association between maternal glucose levels and childhood glucose intolerance and insulin resistance, independent of childhood BMI or genetic predisposition^{60,61}. The Danish National Birth Cohort also found that adolescents born to mothers with GDM had increased adiposity, adverse cardiometabolic profiles and earlier onset of puberty^{62,63}. Intrauterine exposure to

BOX 3

Pharmacotherapy for prevention

Although there are excellent examples of pragmatic approaches to guide therapeutic decision-making^{209,210}, these are in people with T2D. Several pharmacological interventions have shown preventive effects in people with prediabetes, but these findings have not yet been translated into pragmatic frameworks comparable to those used for therapeutic decision-making in established T2D. Nevertheless, for selected individuals at particularly high risk, adding pharmacotherapy to lifestyle-based prevention may help to support a return to normoglycemia when lifestyle changes alone are insufficient to produce durable benefit. Such an approach may be particularly justified in individuals with prediabetes (intermediate hyperglycemia) and established cardiovascular disease, polycystic ovary syndrome, a history of GDM or metabolic dysfunction-associated steatotic liver disease, or in ethnic groups at high risk, including South Asian, Hispanic and South Pacific populations.

The effects of glucose-lowering medications on diabetes prevention are shown in Table 1. In summary, metformin has demonstrated reproducible and sustainable efficacy in delaying progression to T2D in at-risk individuals^{193,211,212} and has been recognized by the ADA Standards of Care for the prevention and delay of T2D as an option in people at very high risk. GLP-1 receptor agonist-based medications, including dual GLP-1 and glucose-dependent insulinotropic polypeptide receptor agonists, are emerging as highly effective pharmacological strategies for diabetes prevention. Nevertheless, debate is intensifying over when, for whom and under what clinical, economic and equity conditions potent pharmacotherapies such as GLP-1 receptor agonists should be added to lifestyle-based diabetes prevention strategies, considering the potential benefits, adverse effects, affordability and equity^{213,214}. This position should be reassessed regularly, particularly in light of the anticipated near-term market entry of lower-cost generics.

Precision therapeutic strategies typically focus on optimizing prediction models for treatment response (or to identify responders and non-responders), yet major challenges for drugs such as tirzepatide, which is highly efficacious for T2D prevention²¹⁵, are the high rates of discontinuation and adverse effects, such as loss of lean mass. Thus, precision prevention approaches might help to optimize medication dosage, combining with multimodal data-informed personalized lifestyle interventions, to mitigate unwanted consequences and maximize the real-world effectiveness of these medications.

pre-existing maternal T2D and GDM combined accounts for approximately 47% of T2D cases in youth, according to the SEARCH case-control study⁶⁴. Additionally, neuroimaging studies suggest that GDM exposure may alter the development of central nervous system regions that are critical for glucose metabolism and energy homeostasis, such as the hypothalamus and hippocampus, which potentially increase an offspring's susceptibility to obesity and insulin resistance^{65–69}.

Follow-up of randomized trials of GDM treatment (usually lifestyle with added pharmacotherapy when required to maintain glycemic targets) has shown clear neonatal benefits on the prevalence of macrosomia but no consistent reduction in childhood obesity or glucose impairment through early adolescence^{70–72}. Some isolated benefits, such as lower FPG in female offspring⁷⁰, have been noted, but overall,

extended follow-up and deeper phenotyping are needed to determine whether specific subgroups of offspring may benefit.

Women with a history of GDM face a sevenfold to tenfold higher risk of developing T2D later in life^{73–75}. Early-onset GDM (diagnosed at <20 weeks of gestation) may confer a particularly elevated risk of T2D, possibly linked to more severe metabolic perturbations^{76,77}. The pathophysiology shows considerable overlap with T2D, involving shared genetic loci^{78,79}, many of which appear to affect β -cell function, which continues to decline after delivery⁸⁰ and interact with pre-existing and/or progressing insulin resistance.

Advances in the machine learning of electronic health records could offer new opportunities for earlier and more precise prediction of GDM, but their utility and cost-effectiveness also depend on data quality, completeness and granularity⁸¹. Other studies have suggested prediction algorithms of T2D after GDM that could help to better target T2D preventive efforts in a very high-risk population of women⁸².

Postpartum and interpregnancy

The postpartum period following a pregnancy complicated by GDM presents a key time for precision intervention in mothers, as well as an opportunity for early identification and monitoring in children exposed to GDM in utero. Meta-analysis of lifestyle interventions (for example, diet and exercise) delivered after pregnancy has shown that structured programs can reduce the incidence of T2D in women with previous GDM, even when postpartum weight loss is modest⁸³. Nevertheless, not all trials have demonstrated clear benefits⁸⁴, illustrating heterogeneity in the interventions and challenges of preventive interventions during this life period when mothers are dealing with many competing demands. However, there is a lack of preventive intervention studies (for example, intensive lifestyle programs) targeting specifically (and at the right time) offspring exposed to hyperglycemia in pregnancy.

Childhood and adolescence

Early-life growth trajectories substantially shape T2D risk. Factors such as low⁸⁵ and high⁸⁶ birthweight, rapid weight gain after age two⁸⁶, a sharp rise in BMI throughout childhood^{87,88} and early onset of obesity with high BMI in youth^{89–94} each increase the risk of IGT and T2D in adolescence and early adulthood. Hormonal changes that occur during puberty may also amplify T2D risk by exacerbating transient insulin resistance⁹⁵. Many adolescents spontaneously return physiologically to normal glucose regulation following puberty; however, for youths with risk factors, such as severe obesity, insulin resistance associated with puberty can trigger rapid progression from IGT to overt T2D^{96–98}. An earlier timing of puberty, particularly early menarche in girls, is consistently associated with a higher risk of T2D in adulthood, independent of adiposity⁹⁹. Modifiable risks for youth-onset T2D include a profile of sedentariness, unhealthy diets and short sleep, which are often underpinned by adverse environments such as food insecurity, family habits and school-based access to unhealthy foods^{100–102}.

Adolescents with T2D experience more rapid β -cell decline¹⁰³ and earlier onset of complications such as hypertension, dyslipidemia, kidney disease and neuropathy¹⁰⁴ than those with adult-onset T2D, underscoring the critical need for early identification of prediabetes (intermediate hyperglycemia) and preventative interventions in this age group. There is an established genetic basis for youth-onset T2D¹⁰⁵, with the risk loci overlapping with those seen in adult T2D¹⁰⁶. This, combined with exposure to GDM⁶¹, contributes to the susceptibility to risk of IGT and T2D in youth. The Physical Activity and Nutrition in Children study showed that a combined physical activity and dietary intervention attenuated the increase in insulin resistance over 2 years in a general population of predominantly normal-weight children¹⁰⁷.

Young adulthood

Key factors in young adulthood that predispose people to developing T2D include an early metabolic drift, often characterized by rapid

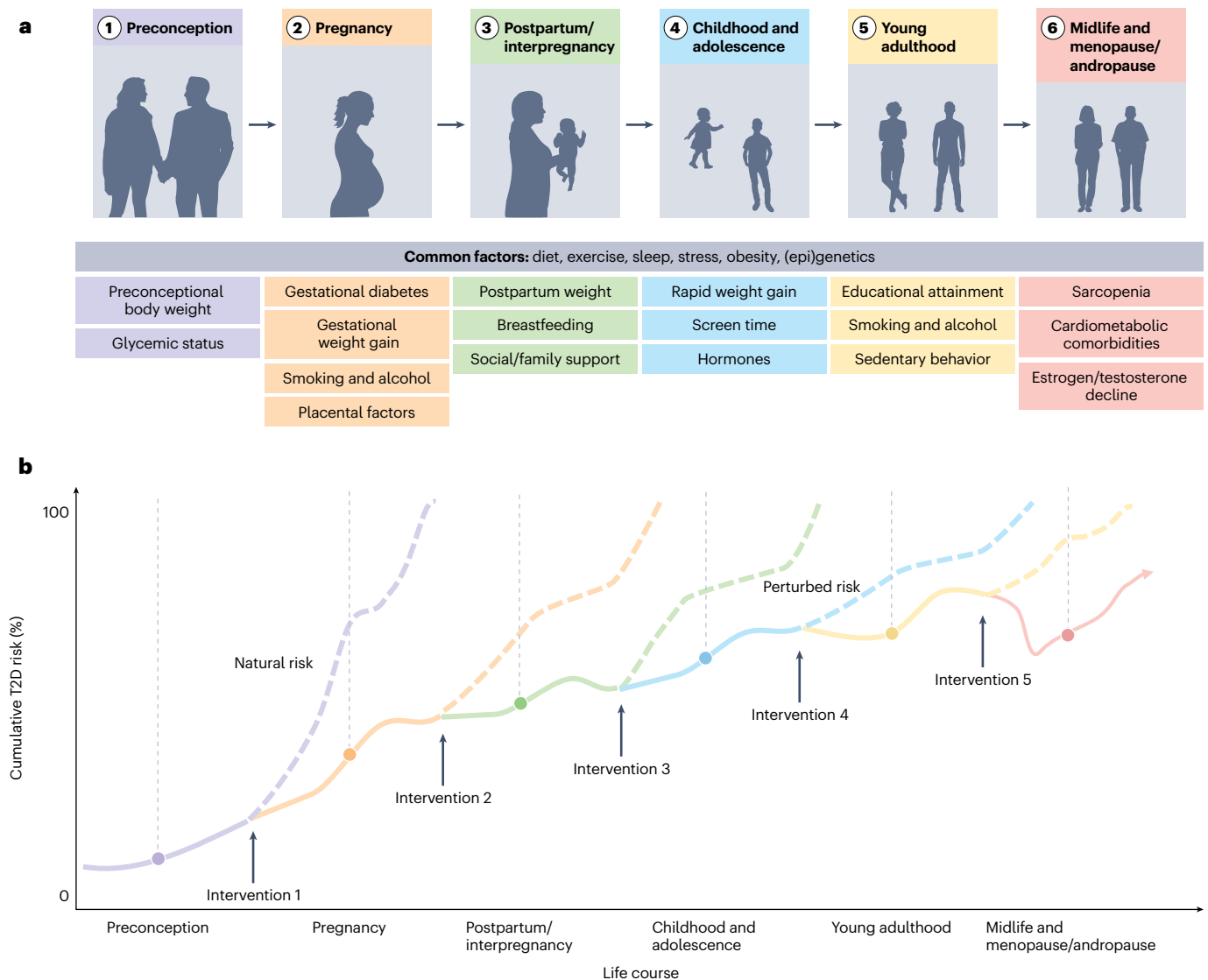


Fig. 2 | Life-course accumulated diabetes risk. T2D risk emerges from cumulative metabolic burden across sensitive periods of the life course. **a**, Critical life stages, including preconception, pregnancy, postpartum, childhood and adolescence, early adulthood, and midlife to menopause/andropause, during which common and specific modifiable risk factors may amplify or

attenuate disease trajectories. **b**, Conceptualization of timely and effective interventions that can attenuate risk accumulation when introduced at different life stages, resulting in durable reductions in lifetime T2D risk; failure to intervene results in progressively perturbed trajectories and higher lifetime risk.

weight gain, weight cycling and difficulty losing weight after pregnancy, which is compounded by modern work and leisure patterns that collectively attribute to increased insulin resistance. Even short-term overeating with calorie-dense, ultra-processed snacks can rapidly increase liver fat and induce disruptions in brain insulin action that persist beyond the period of overconsumption in young, otherwise healthy adults, potentially accelerating this early metabolic drift¹⁰⁸. Furthermore, emerging research indicates that brain insulin action and peripheral insulin sensitivity fluctuate across the menstrual cycle, with insulin sensitivity being higher in the follicular phase^{109,110}.

Midlife and menopause or andropause

During midlife, individuals often face competing demands of balancing career responsibilities, raising children and caring for elderly parents, often resulting in less time for self-care and prevention efforts. Evidence from the DPP showed that adults aged 65 years or older had higher participation rates, better adherence to self-monitoring and greater success in meeting physical activity and weight loss goals compared with younger adults¹¹¹.

During the midlife period, major hormonal transitions create a period of heightened vulnerability for the development of T2D. For women, the menopausal transition brings decreases in estrogen and increases in androgens¹¹², contributing to an unfavorable and pronounced decrease of sex hormone-binding globulin¹¹³ and an increase of visceral adipose tissue^{114–116}. This visceral adipose tissue accumulation is a key driver of insulin resistance and is more strongly linked to metabolic dysfunction in women than in men¹¹⁷. In addition, the EPIC-InterAct study found that early menopause and shorter reproductive lifespan also increase T2D risk^{118,119}. The hormonal and body-composition shifts associated with menopause, together with age-related changes in muscle mass, adipose distribution and insulin sensitivity, may accelerate progression from prediabetes (intermediate hyperglycemia) and may also contribute to end-organ vulnerability before diagnostic thresholds for diabetes are reached. Considering appropriate hormonal therapy and managing menopausal symptoms, especially when initiated early in the postmenopausal period, may improve metabolic profiles and reduce long-term T2D risk¹²⁰.

In men, the evidence for a comparable andropause effect is less definitive. Although a decline in androgens may contribute to unfavorable body composition changes, especially in men with overweight or obesity, who frequently have low serum testosterone concentrations, its independent role in T2D pathogenesis requires further substantiation. Ding et al.¹²¹ indicated that incident T2D is increased in men with low serum testosterone levels, whereas other studies with testosterone supplementation in middle-aged and older men with prediabetes yielded inconclusive results regarding the prevention and remission of T2D^{122,123}.

Precision prevention

The proposed life-course risk architecture, together with the inherent limitations of dichotomous, threshold-based diagnostic criteria, argues for a paradigm shift toward cumulative diabetes risk estimation. A multivariable risk score could quantify an individual's probability of developing T2D within a defined time horizon, analogous to established cardiovascular risk prediction models.

Such tools should be population calibrated and regionally adaptable while allowing dynamic estimation of risk modification in response to lifestyle, behavioral, pharmacological (Box 1 and Table 1) or surgical interventions. This approach would move the field beyond static categorization toward personalized, quantitative risk stratification with direct clinical utility. The German Diabetes Risk Score (drs.dife.de/index.php?lang=en) and Finnish Diabetes Risk Score (findrisc.com/en) are examples of this framework. Advances in prediction modeling further enable the integration of glycemic indices into multivariable algorithms, improving model performance while preserving the continuous nature of metabolic risk.

Precision prevention of T2D goes further than conventional risk prediction, often by accounting for probabilistic risk and estimated treatment response, with interventions designed to ensure the timely delivery of highly effective prevention. Precision approaches should involve modeling both the trajectories of disease progression and the periods when intervening is likely to prove most effective.

Multivariable pathophysiology-based sub-phenotyping in people at high risk of T2D has identified etiological clusters that differ not only in future diabetes risk, but also in rates of complications and mortality risk, even when glycemic deterioration is modest^{124,125}. This emphasizes the need for risk models to go beyond glucose, also accounting for insulin resistance, β -cell dysfunction, body fat distribution, hepatic metabolism, neuroendocrine and hormonal regulation, and inflammation. Nevertheless, despite a diversity of precision analytical methods used to quantify heterogeneity and predict risk and response, there is no clear consensus on which methods to use and how to implement them^{124,126}.

Analyses of deeply phenotyped cohorts have revealed heterogeneity in the causes and consequences of dysglycemia, with considerable heterogeneity in rates of progression from healthy glycemia to prediabetes and then diabetes^{127,128}. Patterns of abnormal postprandial glucose response (PPGR)¹²⁹, an acute and common manifestation of dysglycemia, also differ markedly from one person to the next¹²⁹, influenced by genetic factors, the micro- and macro-nutrient composition of food, gut microbial diversity, sleep and circadian rhythms, meal timing, psychosocial factors and physical activity^{130,131}. PPGR is heavily determined by peripheral insulin sensitivity and β -cell function¹³². Approaches that harness multimodal data have been used to generate signatures that help to identify high-risk individuals (for early prevention and monitoring of disease progression or regression) and target diet or other lifestyle interventions¹³³.

Lifestyle-based precision prevention approaches often use multimodal data derived from nutritional and exercise challenge tests and processed using machine learning methods, as well as digital support tools to design and deliver personalized interventions¹³⁴. Such approaches contrast with standard lifestyle interventions that typically

focus on simple caloric restriction and structured physical activity for 150 minutes per week¹³⁵; the physical activity approach omits the remaining ~93% of waking time that offers opportunities for prevention strategies to incorporate structured and non-structured physical activity and accommodate an individual's circumstances, preferences and capabilities¹³⁶. The PLIS showed that in people at high or low risk of T2D, an intensive compared with regular DPP-like lifestyle intervention was more effective in normalizing glucose tolerance¹³⁷. Secondary analysis from the PLIS highlighted that different clusters of participants (varying by risk factor profiles) respond differently to lifestyle interventions¹³⁸. For example, the insulin-deficient cluster was characterized by improvements in β -cell function. In the severely insulin-resistant cluster, only individuals that lost ectopic liver fat achieved this outcome, emphasizing interactions between the liver and pancreas¹²⁴. Moreover, deviations from an individual's usual sleep pattern and poorer sleep efficiency are associated with exaggerated PPGR the following morning, highlighting sleep as a meaningful interventional opportunity for non-pharmacological metabolic optimization¹³⁹.

Nutrition is complex but represents one of the most mature domains of precision prevention (although a detailed analysis is beyond the scope of this Perspective). For example, a plasma-based multi-lipid score has been developed and validated that not only predicts the development of T2D and cardiovascular disease in people at risk, but also supports an individualized approach to modify a person's cardiovascular risk through diets enriched with unsaturated fatty acids¹⁴⁰. The high between-person variability in PPGR, even when identical meals are consumed^{130,134}, highlights opportunities to personalize dietary recommendations using data from glycemic responses to test meals, sleep patterns, physical activity and, to a lesser degree, the gut microbiome¹³¹. Machine learning models trained on PPGR variability demonstrate that data-driven dietary prescriptions outperform standard dietary guidelines to reduce PPGR in both healthy people¹³⁰ and those with T2D¹³⁰. Importantly, for precision lifestyle approaches to work well in real-world settings, they need to consider the recipient's ability to adhere to diet and other lifestyle recommendations; this requires that a person's psycho-socio-cultural characteristics are, where possible, incorporated into precision prediction models and, ideally, throughout the research and implementation^{134,141}.

Metabolic and bariatric surgeries can each offer powerful glycemic and weight-related benefits, including diabetes and prediabetes remission and disease trajectory modification¹⁴². Various analytical approaches have been proposed to help identify individuals who might benefit from early surgical intervention, such as unsupervised clustering or the prediction of severe insulin resistance or rapid worsening of hyperglycemia¹⁴³. These studies demonstrate greater prediabetes remission in some high-risk clusters¹²⁵, with people within some clusters transitioning to lower-risk states after surgery, unlike those undergoing behavioral modification alone^{144,145}.

Tools for precision approaches

Multimodal data

Genome-wide association studies have identified more than 1,200 loci associated with T2D risk⁷⁹; however, the clinical utility of standard polygenic risk scores for risk prediction is modest compared with traditional clinical factors¹⁴⁶. Nevertheless, genetic data have been successfully used to help differentiate the disease processes underpinning diabetes predisposition using partitioned polygenic scores^{147–149} and in response stratification¹⁵⁰. Moreover, polygenic risk scores are more predictive in younger individuals than in older people¹⁵¹, which may justify their use in youth to help target preventive approaches early in the life course before disease manifests.

Large-scale initiatives, such as updated human plasma proteome atlases and new multimodal proteomic efforts incorporating genetic information (for example, π -HuB), enable more detailed characterization of circulating proteins, proteoforms, extracellular vesicles and

tissue-of-origin signals^{152–154}. Integrating high-resolution proteomic data with cell- and tissue-specific reference maps makes it possible to trace circulating proteins back to their source tissues, providing insights into organ-specific vulnerability, dynamic disease processes and new therapeutic targets for prevention. Metabolomics provides complementary insight by capturing circulating small molecules and pathway activity linked to diabetogenic lifestyle exposures, including branched-chain amino acids, lipids, bile acids, carnitines and amino acid-related pathways^{133,155}. However, both proteomic and metabolomic assays face challenges related to relative rather than absolute quantification and limited standardization across platforms¹⁵⁶.

The integration of multimodal data has become the mainstay of many machine learning analyses in precision prevention research. Popular phenotypic data (for example, data on genes, transcripts, proteins, metabolite variants, clinical phenotypes, behavior and environmental exposure) are frequently derived using rapidly evolving technologies (for example, sequencing, imaging or biometry). When performed well, multimodal analyses can help to determine biological systems-wide processes, identify novel disease endotypes and elucidate molecular pathways that drive disease heterogeneity¹⁵⁷. For precision prevention of T2D, these data may enable the identification of high-risk individuals before dysglycemia becomes clinically apparent, enabling the prediction of an individual's dominant mechanistic pathways and responses to exposures and interventions.

Discordancy analyses

Although obesity correlates with a range of adverse cardiometabolic health characteristics at the population level, not everyone living with obesity experiences these adverse consequences. Understanding the factors that drive biological concordance and discordance between traits may help to map the etiology of obesogenic disease and highlight molecular targets for disease prevention. This concept has been explored with innovative analytical approaches that either juxtapose genetic risk and protective alleles¹⁵⁸ or cluster them¹⁵⁹ to expose novel clinical and molecular features and pathways^{158,160}. The enhanced molecular understanding of cardiometabolic disease that these studies have delivered may aid novel target discovery but is unlikely to be directly translatable into clinical settings for risk or response stratification. By contrast, discordancy analyses with a focus on clinical markers have proven helpful for strengthening established clinical risk prediction models. For example, an analysis of multiple clinical risk markers revealed patterns of discordancy that meaningfully enhanced the prediction and net reclassification of T2D and coronary artery disease beyond the most established clinical risk tool (SCORE2)¹⁶¹.

Digital tools for interventions

In the context of precision prevention, digital health tools encompass more than just measurement devices. They can refer to integrated systems that combine personalized data capture (via wearables such as continuous glucose monitors (CGMs) or physical activity monitors, dietary assessment tools or smartphone apps), complex data processing, integration and storage platforms, and analytical frameworks that transform raw data into actionable, personalized insights^{162,163}. CGM devices and other wearables provide immediate feedback on glycemic responses to food, stress, sleep and physical activity, transforming generic advice into personally relevant knowledge¹⁶⁴. CGM data have revealed marked interindividual variability in PPGRs driven by metabolic, behavioral and molecular factors, enabling machine-learning-based personalization of diets that outperforms standardized dietary guidelines^{129,130}. Beyond prediction, CGM-based digital feedback has been increasingly explored as a behavioral intervention tool in prediabetes (intermediate hyperglycemia) that may enhance adherence to dietary and lifestyle modifications¹⁶⁴. Wearables and apps use goal setting and gamification to enhance adherence, which can help to address the high dropout rates that are common

in traditional interventions¹⁶⁵. At the system level, the progressive digitalization of healthcare has enabled large-scale aggregation of electronic health records, creating opportunities for large-scale modeling of disease trajectories and intervention effects¹⁶².

However, digital tools also have important limitations, including uncertain long-term adherence, device accuracy and reliability, uncertain interpretation of CGM data in individuals without diabetes, interoperability across platforms, digital literacy and privacy concerns^{166–168}. Health inequalities would also widen if access depends on income, education, technology literacy, smartphone availability or insurance coverage¹⁶⁹. Further studies are needed to validate and compare CGMs, wearables and other emerging digital monitoring tools against established clinical standards, to guide the translation of personal data streams into actionable prevention strategies¹⁶³.

Overall, continuous individual-level monitoring coupled with machine learning may reveal complex interactions between biological and environmental features that signify early divergence from normoglycemic trajectories, predict individual-level tipping points where glycemic deterioration begins to accelerate rapidly and help to forecast the best timing and format of preventative interventions.

Barriers to prevention

Despite extensive knowledge about the etiology of T2D, it remains unclear why diabetes prevention is so difficult to achieve in people at increased risk. A constellation of obesogenic environmental features have been identified, including nutrient-dense, ultra-processed foods, automation of daily life leading to reductions in physical activity, sleep disturbance, excessive stress and many more^{170,171}. There are also several key cultural, behavioral and clinical factors that contribute to a lack of effective contemporary prevention⁴⁴. A further barrier is individual engagement and acceptance of prevention interventions. Prediabetes (intermediate hyperglycemia) and early metabolic risk are typically asymptomatic, and individuals may not perceive non-diabetes as warranting active management.

Prediabetes (intermediate hyperglycemia) is often diagnosed late in the trajectory to T2D, when substantial β -cell failure has already occurred¹⁷², making lifestyle intervention to prevent progression or achieve remission challenging. We propose a requirement for the International Diabetes Federation (IDF) granular staging schema involving 1 h plasma glucose (Box 1), in addition to traditional criteria, to appropriately classify the degree of risk and determine adequate therapeutic recommendations. Furthermore, many individuals with dysglycemia are unaware of their condition¹⁷³ or are aware but do not know that intervention is appropriate given that they do not meet specific thresholds for prediabetes (intermediate hyperglycemia) or diabetes.

In many countries, national agencies and healthcare systems have not prioritized funding for the prevention and control of non-communicable disease, with national diabetes programs such as the US DPP and Finland's FIN-D2D facing funding cuts or closure after a limited period of availability¹⁷⁴. In settings where such programs do exist, deficiencies in training and education for healthcare providers result in a lack of awareness of programs¹⁷⁵ and consequent low referral rates for individuals at high risk of developing T2D. The lack of integration of such programs into already fragmented healthcare systems that emphasize treatment over prevention interventions also contributes to the rising prevalence of T2D. These barriers are amplified by the fact that intensive prevention requires trained personnel, sustained engagement and upfront investment, whereas health systems must allocate limited resources across many competing disease areas. Prevention strategies in high-risk individuals are generally considered cost-effective^{176–178}; however, economic evidence specifically using prediabetes remission as the primary outcome remains limited. A health-economic evaluation from the PLIS suggested that risk-stratified lifestyle intervention can be cost-effective for achieving prediabetes remission in lifetime modeling¹⁷⁹. This modeling suggested potential

cost savings, although short-term quality-adjusted life-year gains were not demonstrated, which is plausible because upfront costs and efforts may only translate into benefits over longer time horizons. Yet, screening costs, the number of individuals who need to be identified and treated to prevent clinically meaningful outcomes, and real-world implementation requirements remain important uncertainties. Thus, prediabetes remission is a promising health-system goal, but implementation will require pragmatic risk-stratification tools, affordable delivery models, adequate workforce capacity and engagement in communities outside the standard healthcare systems.

On a global scale, the burden of T2D and undiagnosed and untreated T2D is increasingly borne by low- and middle-income countries¹⁸⁰. In addition, people at high risk and with prediabetes (intermediate hyperglycemia) are often not detected. This inequity is driven by numerous factors, including commercial determinants of health that drive an increasing global prevalence of adiposity, biological factors that confer increased risk of disease in specific ancestral populations and a lack of access to new-generation therapeutics, which are effective for metabolic control and obesity management but unaffordable in many settings (including disparities in some high-income countries). In the USA, Black, Hispanic and Native American populations, as well as groups of low socioeconomic status, are at increased risk of prediabetes (intermediate hyperglycemia) and T2D, largely owing to social, commercial and environmental determinants of health⁴⁴, as well as a multitude of structural barriers to accessing preventative healthcare.

Barriers to access for therapeutics and diabetes prevention programs are not the only drivers of inequity for the prevention of T2D. Much of the evidence base that informs global and regional diabetes policies and treatment guidelines has been generated in high-income settings with populations of European-derived ancestries¹⁸¹. In the case of prediabetes, a majority of research findings have been derived from white populations characterized by high adiposity and insulin resistance, whereas in regions such as India, where there are an estimated 136 million people living with prediabetes¹⁸², the condition is associated with IFG rather than IGT¹⁸³. In those with IFG, the prevention of diabetes by lifestyle has been shown to be less effective¹⁸⁴. In these and other non-European ancestral groups, the relationship between adiposity and prediabetes or T2D may differ by body fat distribution, ectopic fat deposition, insulin resistance, β -cell function and other pathophysiological pathways that contribute to heterogeneous risk^{185–188}. These differences reinforce the need for scalable precision prevention that is not only biologically appropriate but also affordable and adaptable across socioeconomic and health-system contexts.

Future outlook

Recognizing prediabetes (intermediate hyperglycemia) as a dynamic state along a continuum, future staging criteria will need to shift away from complication-based classifications to pathophysiology-driven frameworks. A staging model for T2D, analogous to that proposed for type 1 diabetes¹⁸⁹, would enable the classification of distinct disease phases and pathophysiological drivers before dysglycemia emerges while recognizing that individuals reach similar functional stages via heterogeneous and nonlinear mechanisms. This could facilitate earlier intervention, more precise risk stratification and a biologically coherent definition of disease progression.

Risk is not constant across life; vulnerability clusters in windows of heightened biological plasticity and susceptibility. Screening strategies should therefore incorporate weight trajectories, gestational dysglycemia, family history and emerging biomarkers, rather than relying solely on age-based criteria or BMI.

This shift also means that prevention should not wait for abnormal glucose values. Metabolic resilience may erode silently for years before glycemic thresholds are crossed, and interventions to preserve function may be more effective than attempts to reverse

dysglycemia once β -cell failure is more advanced, especially early in life. Advances in health technologies that generate high-resolution phenotyping, together with AI systems capable of integrating complex and dynamic life-course data, may allow more precise characterization of metabolic trajectories and detection of modifiable inflection points. Legislative and regulatory efforts to modernize diabetes prevention should therefore support standardized, stage-based terminology. Defining categories such as early-stage, presymptomatic and symptomatic T2D would strengthen screening policies and create accountability, shifting risk profiles toward lower-risk stages and metabolic health.

Within this framework, remission should become a central end-point of T2D prevention, but it must be durable and biologically meaningful, rather than a transient normalization of glucose values. This will require robust predictive markers to identify who can achieve remission, who is unlikely to respond and who is at risk of relapse. Risk-stratified intervention studies such as the PLIS support this principle by showing that intervention intensity and metabolic phenotype influence remission to normoglycemia. Importantly, emerging PLIS health-economic analyses are consistent with the potential value of a risk-stratified prevention strategy while also underscoring that broad implementation will require affordable delivery models and adequate workforce capacity.

Precision prevention should also define the role of pharmacotherapy. With increasingly potent incretin-based and potential emerging disease-modifying therapies, the question is no longer whether pharmacological approaches can delay progression from dysglycemia to T2D, but when, in whom and for how long they should be deployed. Lifestyle, behavioral and pharmacological approaches should therefore be viewed as complementary tools whose use depends on disease stage, mechanism, likelihood of remission, relapse risk, cost and acceptability.

Finally, equity must be embedded across this entire agenda. Precision prevention will fail if it becomes a privilege-based concept available only to well-resourced populations. Across all settings, prevention policies should be tailored to individuals and communities at increased risk of diabetes, with their voices at the center of prevention design. On this basis, we identified ten priorities for a new agenda in T2D prevention, linking biological staging, life-course screening, mechanistic heterogeneity, durable remission, dynamic precision tools, pharmacotherapy, hard outcomes, scalability and equity (Box 2). Together, these priorities shift the focus from identifying risk to changing trajectories toward the goal of sustainable metabolic health.

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their respective areas, critically reviewed and further developed the content and concepts, and reviewed and edited the manuscript. All authors approved the final version of the manuscript.

Competing interests

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